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Bio-Circular Materials in Regenerative Design: design's role in ecological and societal transitions

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Today's social, political, and environmental emergencies require specific efforts in terms of thinking and acting in design. Regenerative design becomes a way to think and act beyond the human-centred boundaries, adapting to more-than-human knowledge and ways of being in the world. Bio-circularity provides a critical framework that can contribute to society and be utilised by policymakers, authorities, industry, NGOs, and academia. When designing for bio-circularity, several key principles and criteria should be considered and integrated into the design process. Designing out waste and pollution is an overarching principle that can be identified in most frameworks or guidelines in the literature (Ellen MacArthur Foundation, 2024), and substitute non-renewable materials with biological alternatives (Holden et al., 2023; Marshall et al., 2023). The primary feedstock should be of sustainable biological origin, and it is important to ensure that it is not extracted from a carbon sink, such as coal or peat (Holden et al., 2023). Prioritising local feedstocks contributes to achieving industrial symbiosis and the possibility of designing for closed-loop systems. In the bioeconomy, renewable biological resources are converted into food, feed, bio-based products, and/ or bioenergy via innovative, efficient technologies (bioeconomyalliance.eu, n.d.). In bio-circular design, materials are selected for their sensorial qualities, technical performance and ability to return to nature as nutrition.

The conversion to a circular bio-based economy means a transition from an economy that, to a large extent, has been based on fossil fuels and growth at all costs to a resource-efficient economy based on renewable raw materials that are produced through the sustainable use of ecosystem services from land and water (FORMAS, 2012). A thriving bioeconomy is centred around people and the natural environment, and it is crucial to move beyond fossil-feedstock dependence and toward a circular, carbon-neutral economy. Strategies for fostering more species-rich systems in agriculture, aquaculture, and forestry, as well as protecting large contiguous biodiverse systems to prevent species extinction and the erosion of biodiversity, becomes important (Palahí et al., 2020). Managing the land sector (agriculture, forestry, wetlands, and bioenergy) sustainably and holistically could contribute up to 30% of the global climate mitigation effort, while simultaneously addressing urgent health challenges. In the transition towards a circular bioeconomy, a conflict of interest often arises when the primary crop or nutritious byproducts from the food industry are primarily used for the energy sector, even if they have the potential to be utilised for other purposes, such as feedstock, biochemicals, and pharmaceuticals. However, the Renewable Carbon Initiative (2025) states that this dual use ensures that the critical nutritional components, such as proteins, are supplied to the food system, while carbohydrates and oils are directed towards industrial uses. The circular bioeconomy represents a crucial step towards transforming our land, food, health, and industrial systems. A circular bioeconomy requires change across multiple sectors, and engagement across these sectors will be the key to success. The UN reports that 20 out of 143 countries

have both circular and bioeconomy strategies, but only Estonia and the Netherlands have integrated them so far (UNCTAD, 2025). Approaches need to be situated and designed for the local context, as it is connected to the sustainable use of ecosystem services. The Global North and Global South face different challenges that the circular bioeconomy needs to address. The bio-circular economy can provide the framework for a just transition to regenerative systems, as it holds the potential to transform the economic and social fabric of communities and regions. The ability to shift crops between the food, feed, and industrial markets enables farmers and industry sectors to respond swiftly to changes in demand and mitigate the risks associated with climate change (Renewable Carbon Initiative, 2025).

A pilot study was conducted to generate knowledge as a first step toward developing a process to guide regional actors toward becoming bio-circular. The study is situated within a regional food system, and the participants included a design researcher, design students, companies from a regional organisation, a circular development hub, and local government. Five companies were selected, all members of the regional food industry organisation, and had participated in different regional network activities aimed at stimulating a circular transition. One company was a start-up from academia, two companies were family-owned SMEs, and the last two were larger companies. The next step was to identify by-products that were considered waste at the time. During the selection process, a new criterion was added because some by-products could be used for food production if novel food-technology methods were available. The start-up from academia, The Plant Protein Factory (*The Plant Protein Factory,* 2025) has developed a process to extract green proteins and other nutrients from green mass for use in food and feedstock. For the project, residues from the extraction of kale, originating from the local food industry, consisting of cut-offs, stems, and discarded leaves, as well as sugar beet tops, were selected. The other by-products originated from local small-scale chocolate manufacturing, industrial hemp processing, brewer's spent grain from a beer brewery, and residues from pressing linseed and rapeseed oil. Data was, in some cases, available for the chemical composition and nutrients. In the other cases, a literature study was conducted to find the information. The by-products were introduced to the students, and they also visited the production facilities to gain a deeper understanding of the processes. During a five-week course in Material-Driven Design (Mau, 2020), the B.A. students tinkered, developed material samples, and conducted tests to identify the technical properties and sensorial characteristics of the materials. The only additives available during the experiments were bicarbonate, vinegar (24%), rapeseed oil, and glycerol. Students were encouraged to use as little water and energy as possible. Finally, they developed prototypes that conveyed the material's qualities and potential areas of application. The results were presented to the participants in an exhibition. The material samples and prototypes facilitate ongoing discussion, serving as tangible examples of bio-circular potential in local by-products and of how they can be identified as feedstocks at an early stage.

Kale is a traditional crop grown in fields in the region of Scania and Halland, in Sweden. Kale, like its cabbage relatives, is a crop that requires minimal land and is climate-resilient in northern Europe, as it can withstand frost and snow. Kale contributes to lowering the climate impact of the food system, and it is mainly consumed locally. After The Plant Protein Factory extracts the proteins, fibres composed of cellulose remain that can be utilised in materials. The teams focused on different potentials of the components in the kale residue and applied two different strategies: (1) the coarse fibres and how they could be turned into fibreboards, (2) the potential of cellulose in the green mass, and investigated its application in papermaking. The first strategy resulted in a lightweight fibreboard that could be moulded into 3D shapes. The final concept was a screen divider for offices, as the material has a pleasant hay-like scent and potentially offers acoustic properties. The second strategy yielded a paper with textile properties: a smooth, flexible material that

is difficult to tear and can be stitched, with a logo that can be embossed into the material. These materials serve as tangible examples of how the food system can be connected to the manufacturing of situated biocircular materials. Before developing these materials, there was an exploratory conversation with the regional packaging industry on how excess cellulose from the food system could be utilised as materials. When introducing physical material samples for the packaging industry, the conversation shifted from '*being an interesting idea'* to '*we need to explore this possibility'*. Facilitating conversation through physical, tangible samples, biocircular materials and regenerative design offers lenses through which industry can explore and imagine possible futures. Practices of making and imagining can engage actors and encourage them to navigate uncertain times and unforeseen challenges.

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